# Lymphoma Image Classification Using Popular Neural Networks

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4/22/2023

Springboard Capstone 3

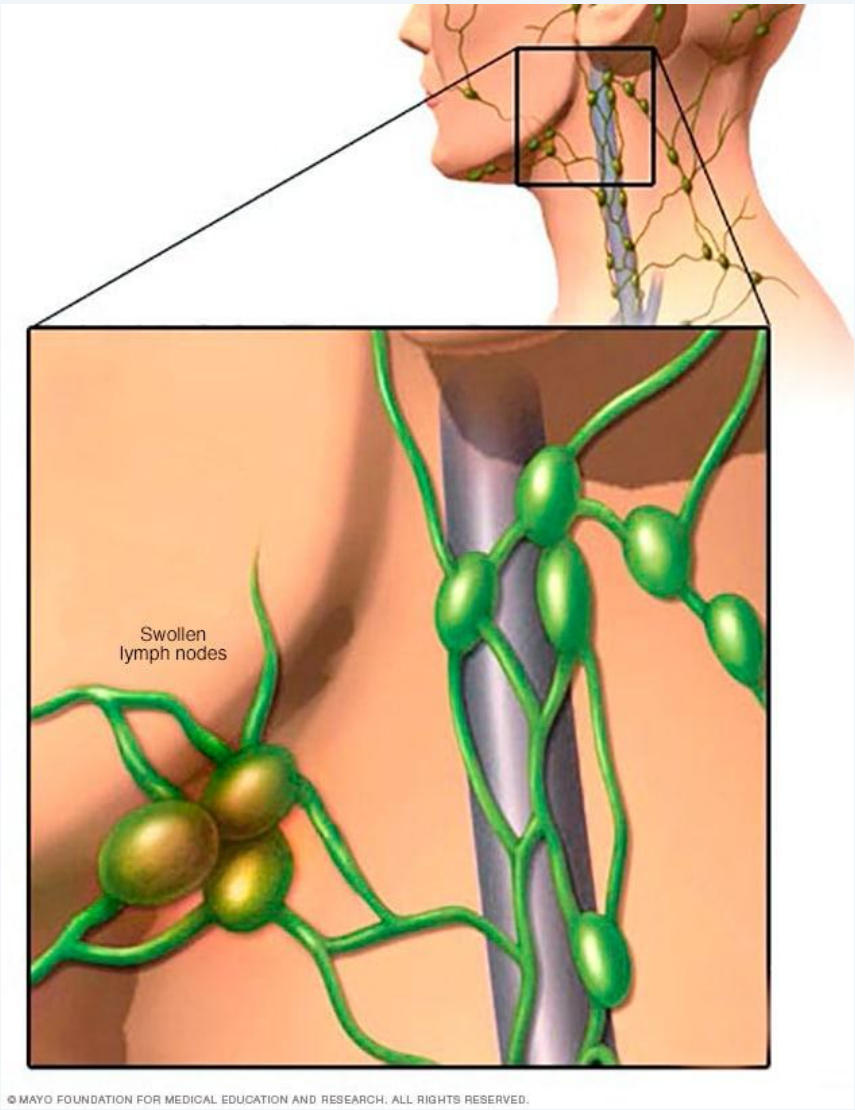
# Agenda

- Background
  - Lymphoma
  - Cancer image identification
- Data Cleanup
  - Brightness
  - Flips/Rotations
- ResNet-50
- Virtual Image Transformer (ViT)
- Model Comparison
- Conclusions and Next Steps

# Lymphoma

- Lymphoma is one of the most common cancers in the USA (4% of all cases)
  - Roughly 80,000 people are expected to be diagnosed with Non-Hodgkins Lymphoma this year
- Diagnostics of Lymphoma may include the following
  - Physical Examination
  - Blood Tests
  - Imaging tests
  - **Biopsy**
    - **Lymph node tissue is viewed under a microscope**

**Example of Lymphoma**



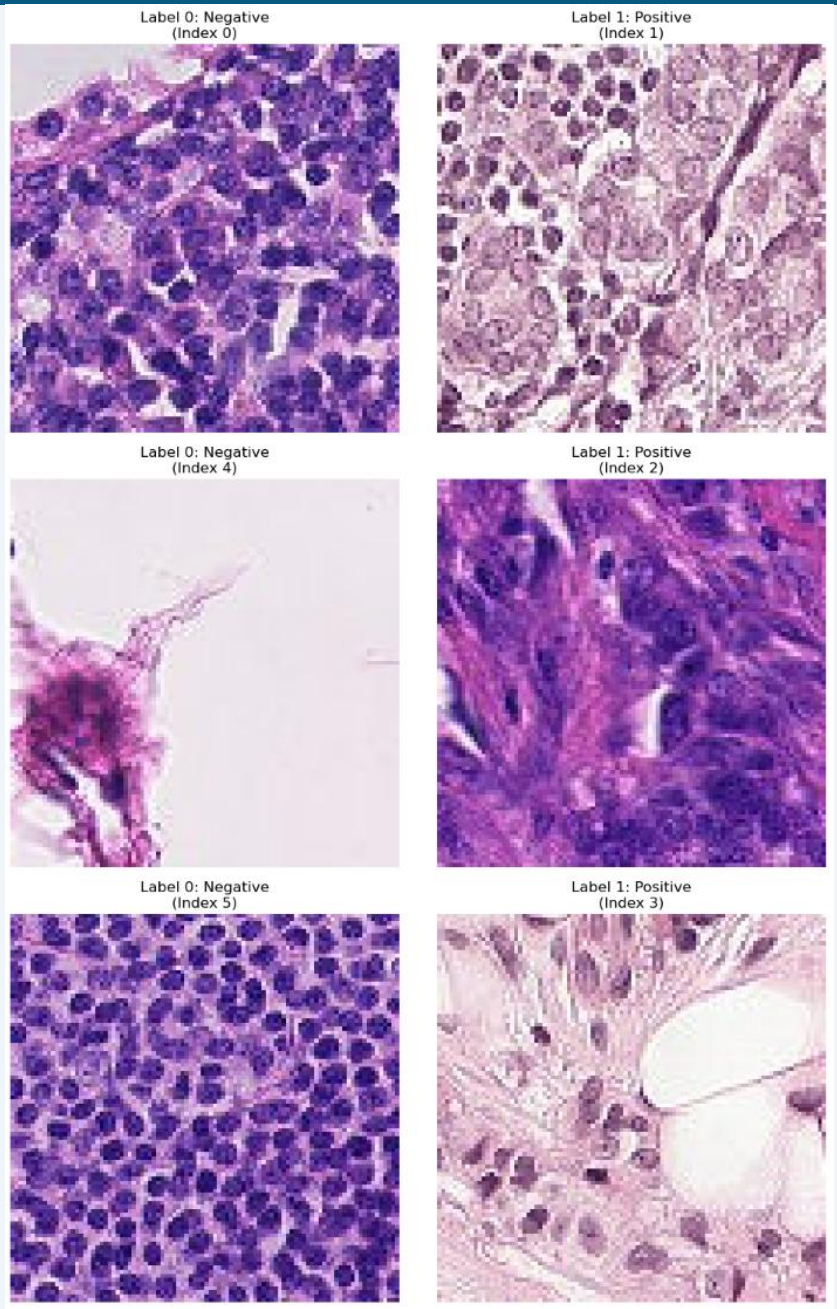

Images and information are from: How Common is Lymphoma? | Key Statistics for Non-Hodgkin Lymphoma | American Cancer Society
Lymphoma - Symptoms and causes - Mayo Clinic

# Classifying Images as Cancer

- One of the main methods for diagnosis of lymphoma is looking at histology slides of cells from a biopsy.

- Common traits of Lymphoma cells:
  - Large irregularly shaped cells
  - Large or irregular nuclei

- Identification of lymphoma through images may be difficult and slow, so ML/AI could be used to improve our ability to find these cells



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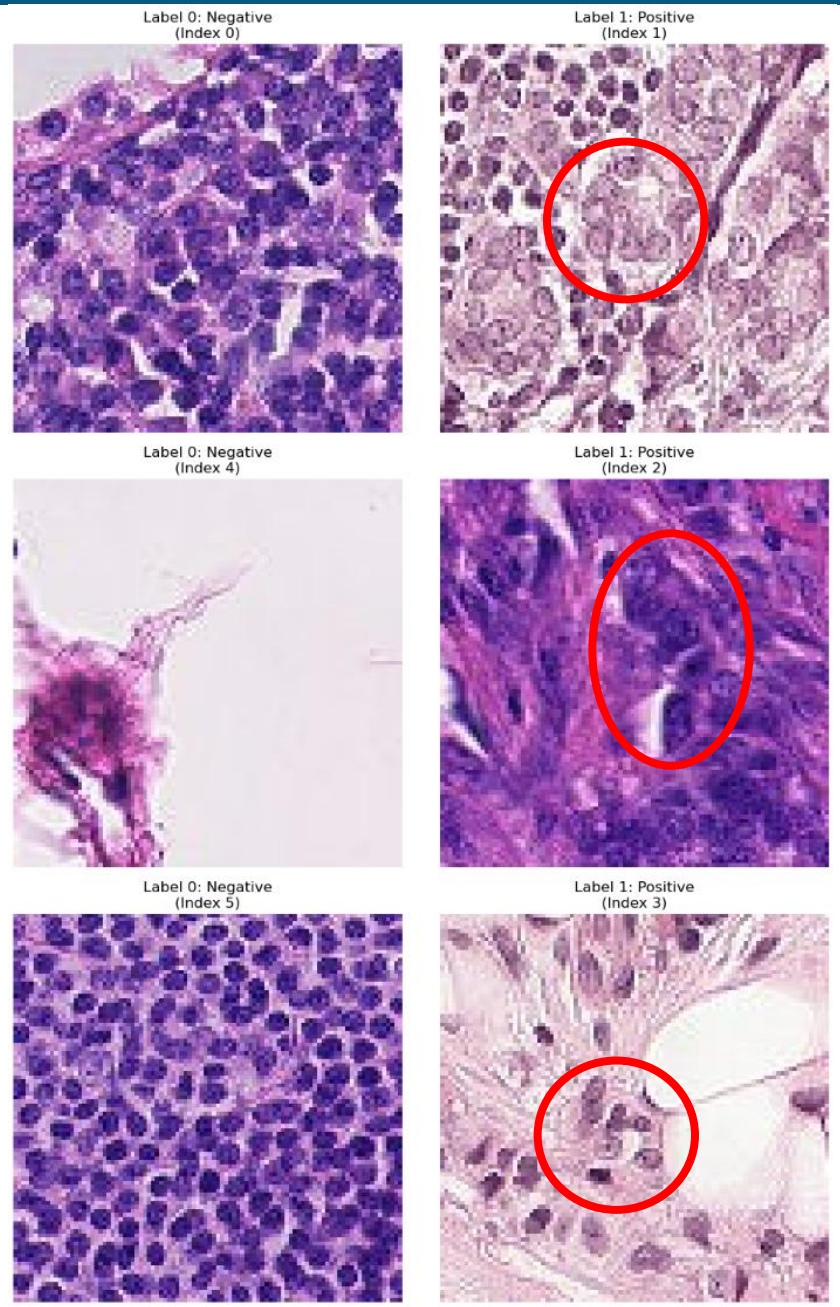

# Cancer Classification Metrics for Success

- Cancer Identified (True Positive):
  - Cancer can be treated
- Cancer is not found (True Negative):
  - Healthy patient can go home and not worry.
- If cancer is identified where there is no cancer (False Positive):
  - Patient is checked further
  - Will be costly for extra tests and potential for unnecessary treatment
- If cancer is not identified (False Negative):
  - Patient is cleared
  - Cancer may progress unknowingly and potentially killing the patient

| Metric | General Description |
|---|---|
| Accuracy | Overall correctness |
| Recall | True Positives to False Negatives |
| Precision | Relating True Positives to False Positives |
| F1 | Combination of Recall and Precision |

$$F1 = 2 * \frac{(Precision \ * \ Recall)}{Precision + Recall}$$

# Cancer Classification Metrics for Success

Priority

1

- Cancer Identified (True Positive):
  - Cancer can be treated

2

- Cancer is not found (True Negative):
  - Healthy patient can go home and not worry.

2

- If cancer is identified where there is no cancer (False Positive):
  - Patient is checked further
  - Will be costly for extra tests and potential for unnecessary treatment

1

- If cancer is not identified (False Negative):
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  - Cancer may progress unknowingly and potentially killing the patient

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# Cancer Classification Metrics for Success

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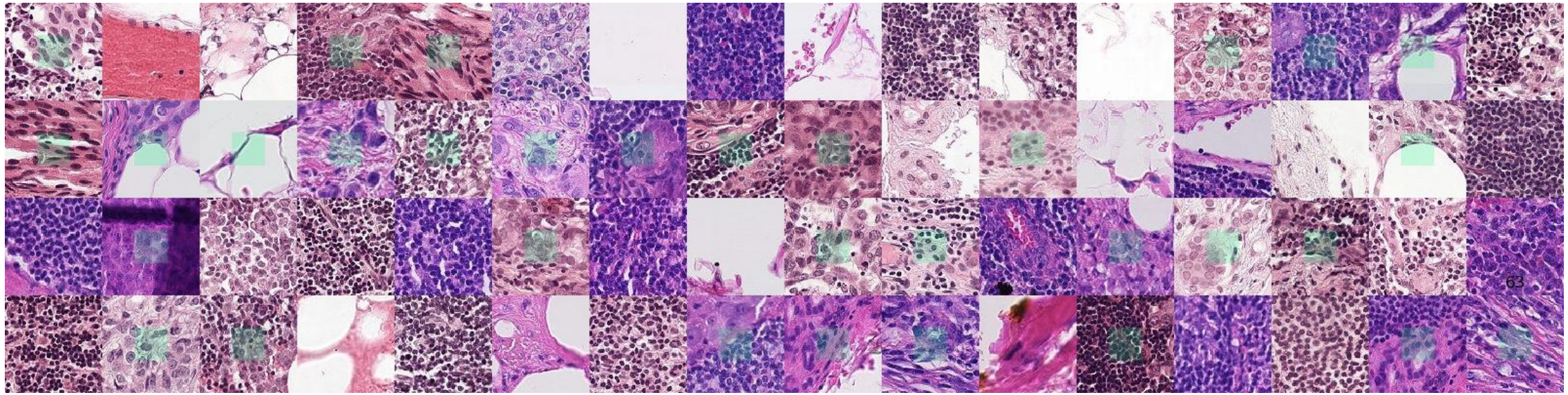
| Priority | Metric | General Description |
|---|---|---|
| 2 | Accuracy | Overall correctness |
| 1 | Recall | True Positives to False Negatives |
| 2 | Precision | Relating True Positives to False Positives |
| 2 | F1 | Combination of Recall and Precision |

$$F1 = 2 * \frac{(Precision \ * \ Recall)}{Precision + Recall}$$

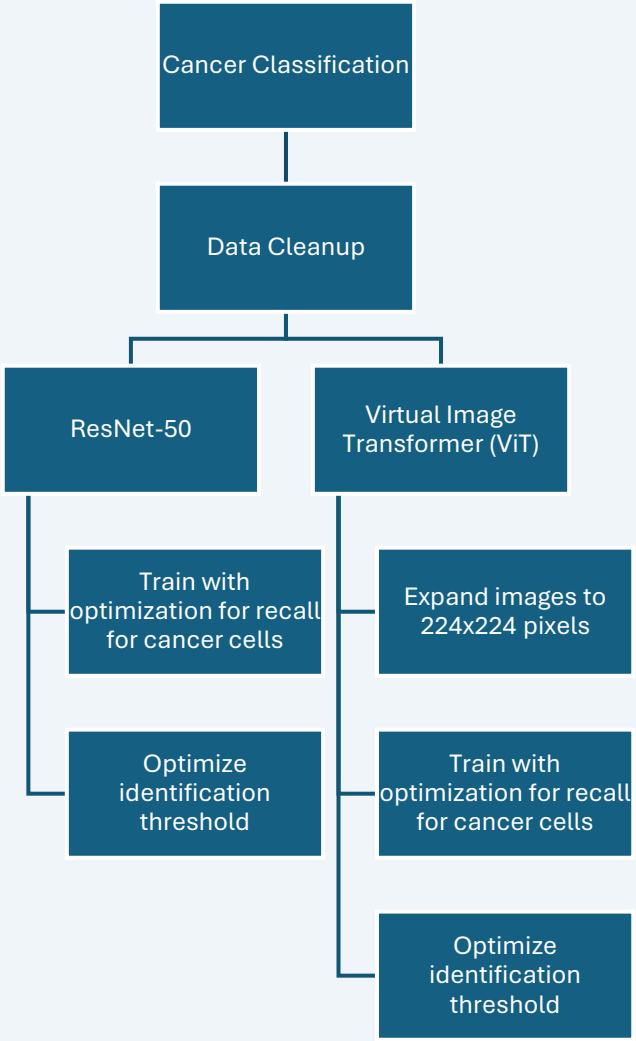
# Image Dataset Description



- Patch Camelon dataset:
  - [GitHub - basveeling/pcam: The PatchCamelyon (PCam) deep learning classification benchmark. · GitHub]
  - 262,144 images in the training set (96x96 pixels)
  - 32,768 in the testing set (96x96 pixels)
  - Data is split 50/50 cancerous and noncancerous

# General Study Plan

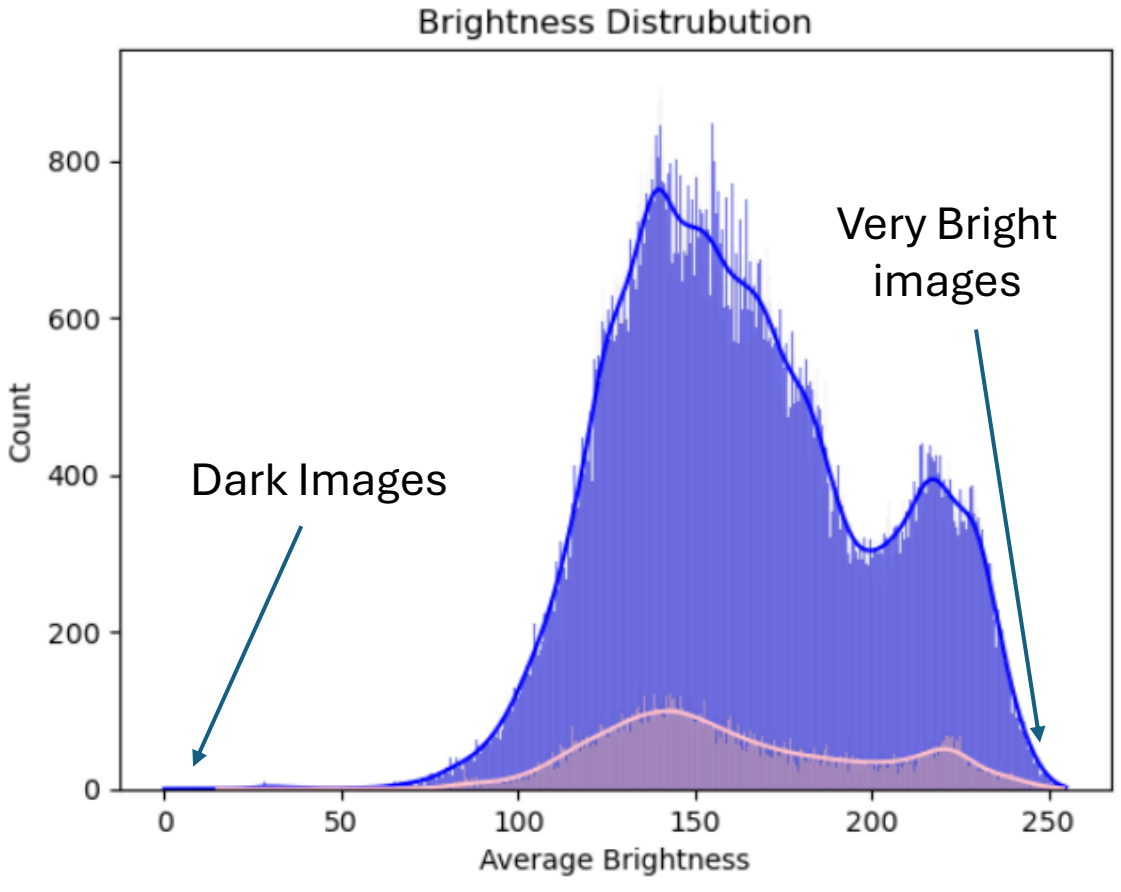
- Focus on Cancer Recall
- Data Cleanup will consist of image transformations and brightness
- Two models were chosen for modeling
  - ResNet-50
    - 50 layers of a popular image classifier neural network
      - Briefly, it works by identifying sharp contrasts withing segments of the image
  - Virtual Image Transformer (ViT)
    - Inspired by electrical transformers
      - Briefly, it works by comparing the whole image against other parts of the image



# Initial Image Exploration: Brightness

- Brightness was calculated by adding the total values for the red, green and blue channels.
  - 0 – Black
  - 255 - White

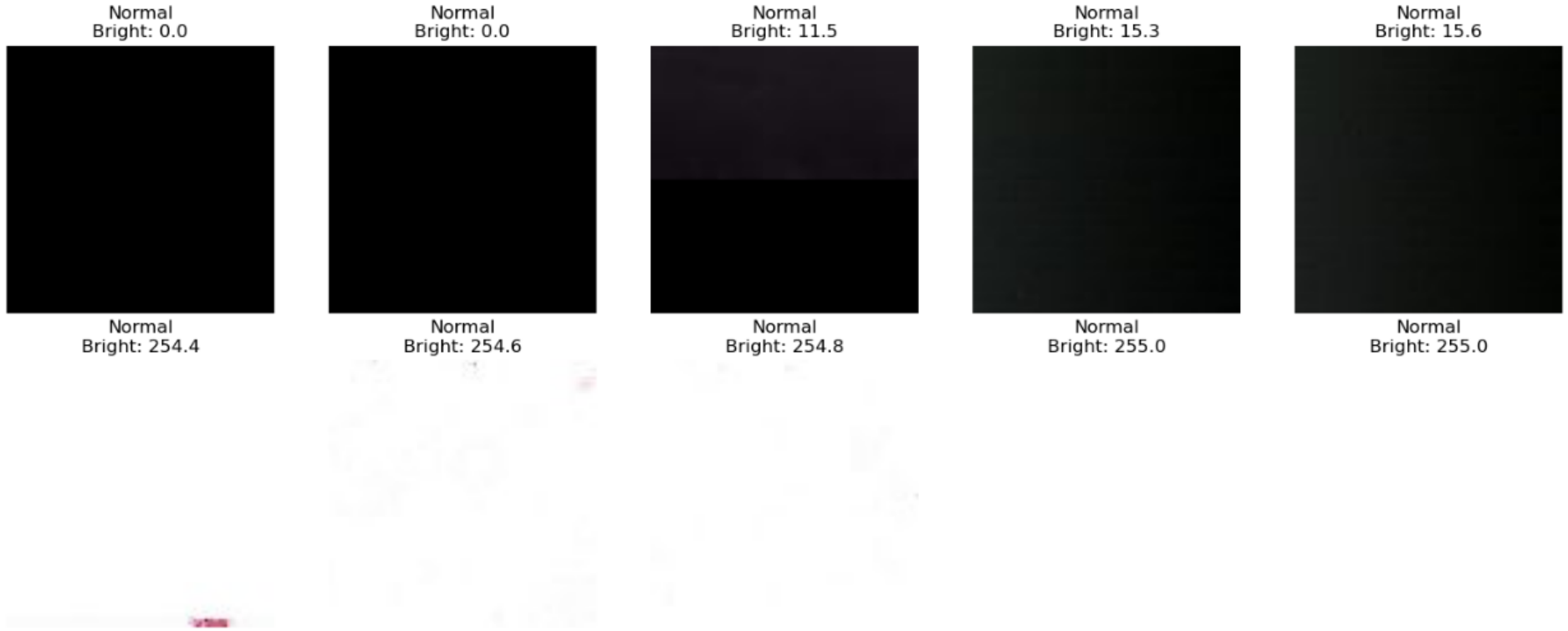
- Images at the edges of brightness are likely only black or white images



# The Brightest and Darkest Images

Darkest Images

Lightest Images

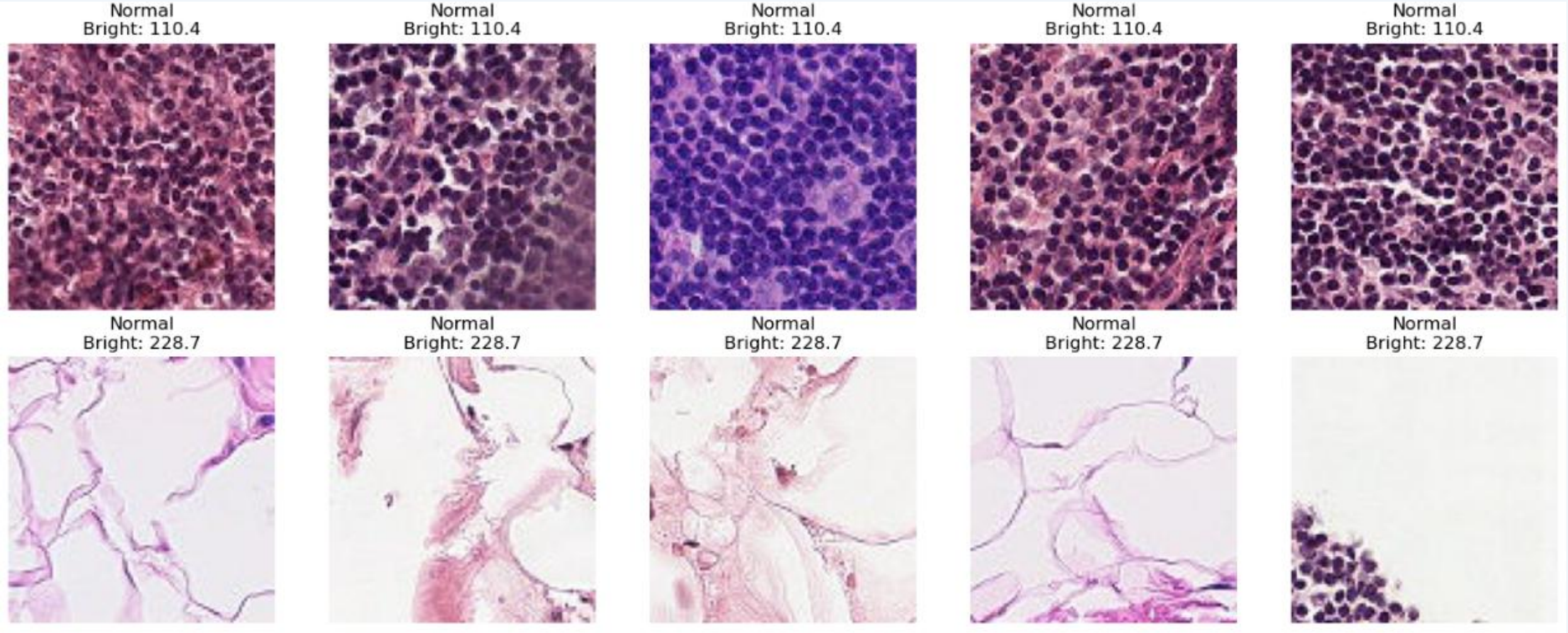



- It is not possible for anyone to determine if these images show cancer or not.
- The bottom and top 5% of images by brightness were removed

# The Remaining Lightest and Darkest Images

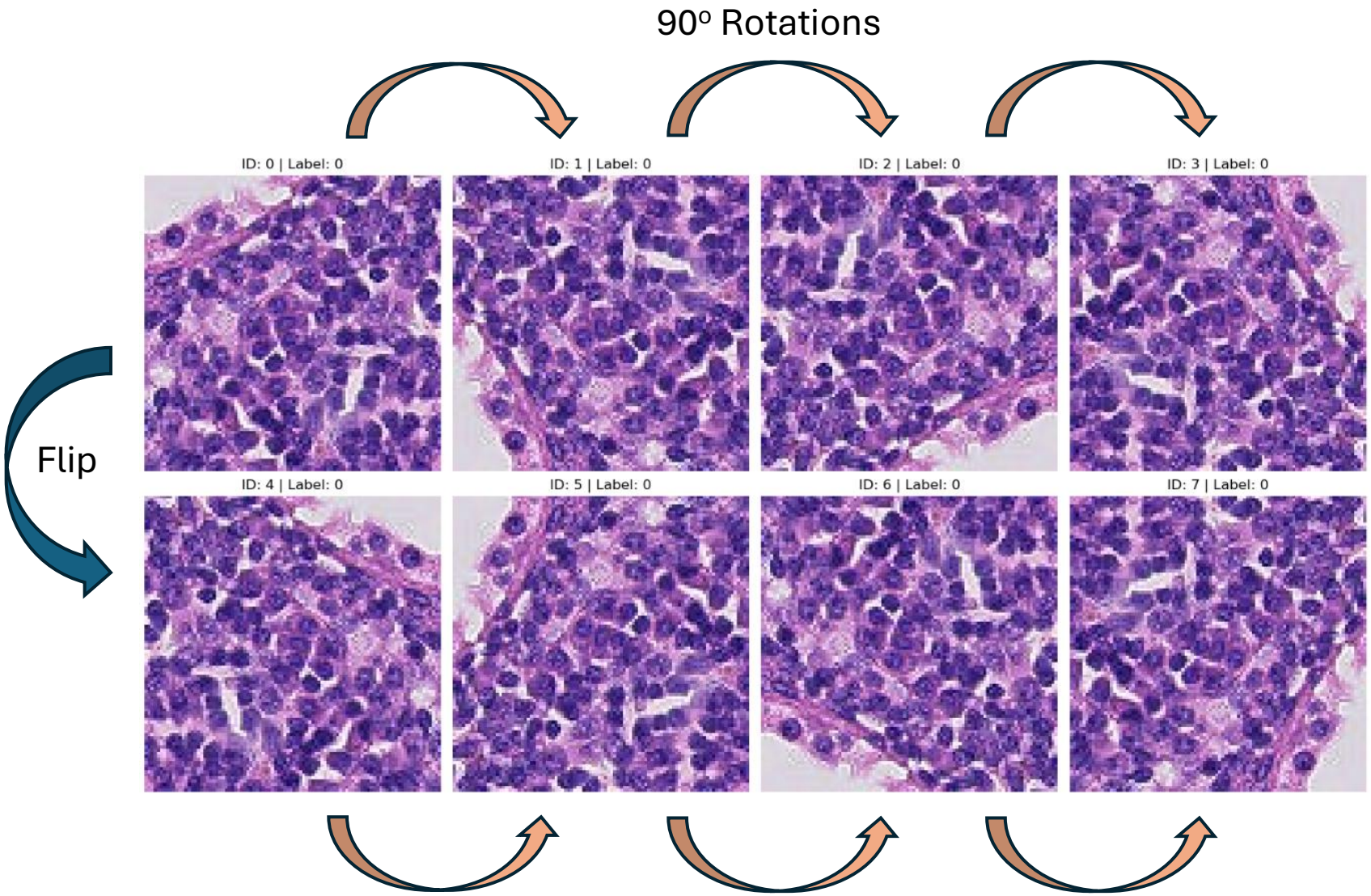
Darkest Images

Lightest Images



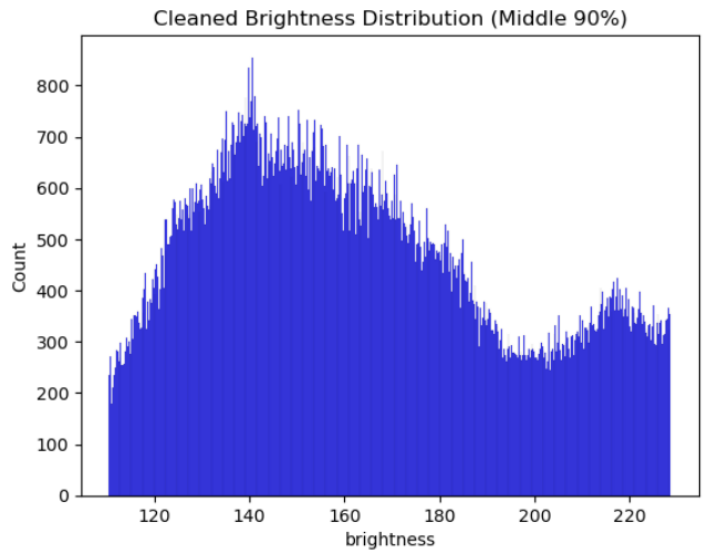

- After removal of the most/least bright images, 90% of the samples remain
- 235893/262103 images remain for model training

# Image Rotation and Flipping for Generalization




- All training images were flipped and rotated
  - Generalizes data
  - Helps prevent model overfitting

# Final Image Descriptions




- A total of 1,887,144 million images can be used for training models
  - After brightness trimming
  - After image rotations

- Test dataset includes 29490 images
  - After brightness trimming
    - Wanted to keep testing images within brightness limitations of the model
  - Images not rotated
    - Did not want to alter the images in any way

- This image set will be used to compare two popular image recognition neural networks

| Metric | Training Dataset | Test Dataset |
|---|---|---|
| Sample Count | 1887144* | 29490 |
| Array Shape | (1887144,96,96,3) | (29490,96,96,3) |
| Data Type | Uint8 | uint8 |
| Min Pixel Brightness | 110.4 | 110.4 |
| Max Brightness | 228.7 | 228.7 |
| Tumor % | 53.92% | 53.40% |

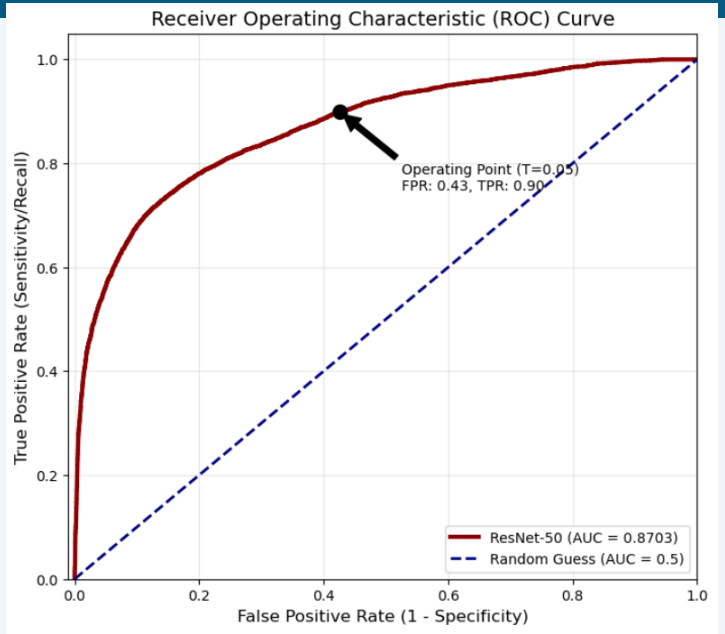
* Original training set was 235893 images (after image removal from brightness, but were rotated or flipped for a total of 8X images.

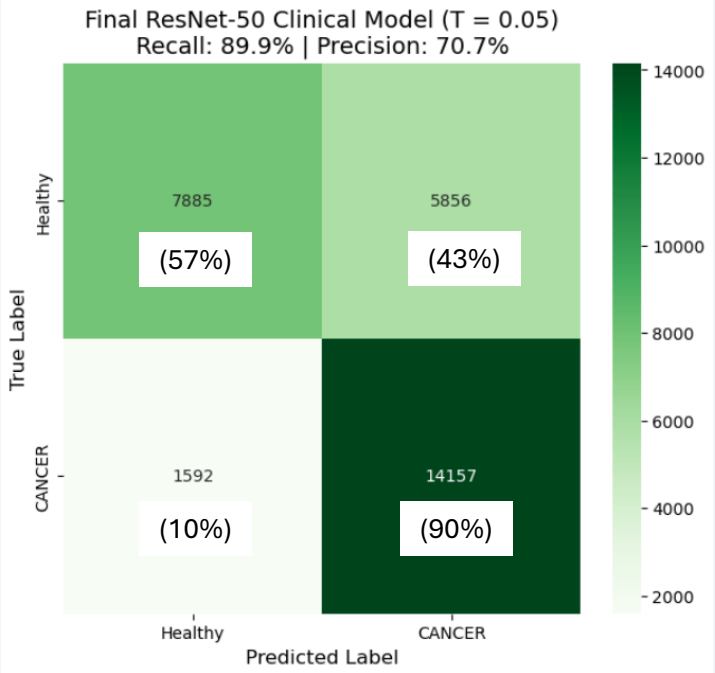
# ResNet-50

# ResNet-50: Final Model Performance

- The Cancer Recall was significantly improved to 90%
  - 43% of the healthy images were identified as cancerous – may lead to unnecessary treatment/testing

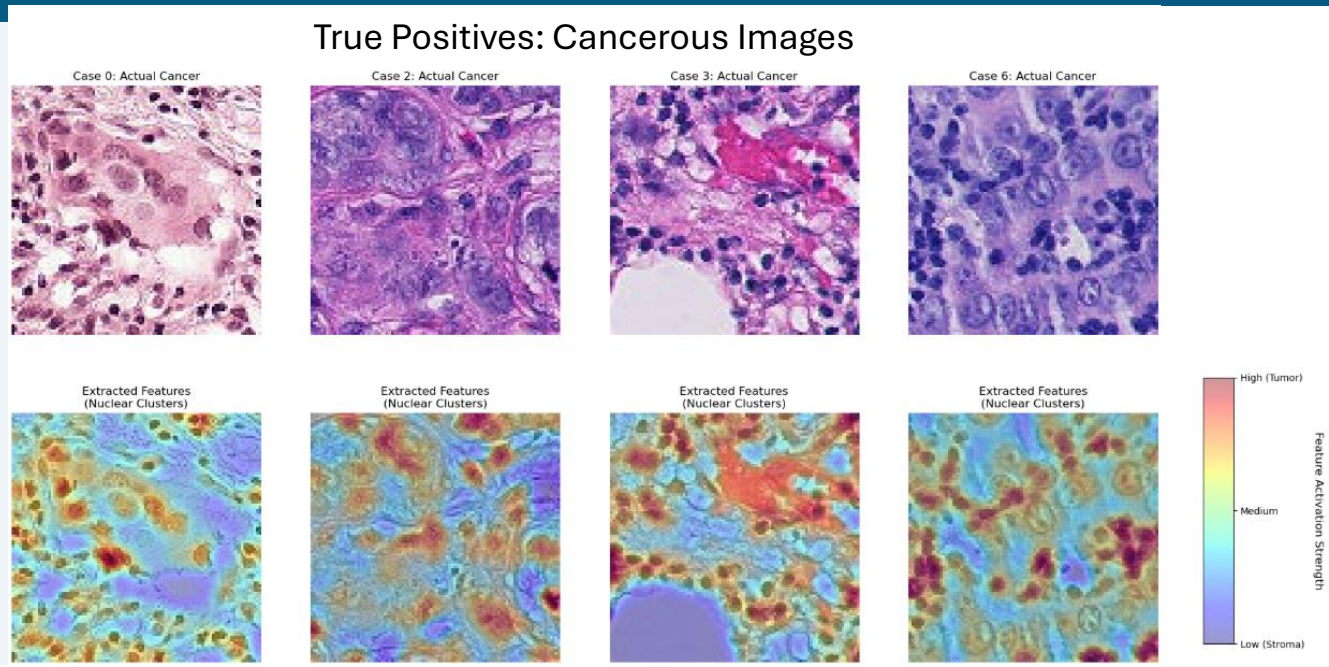
- This model struggles with precision, but can identify most cancer images which was the top priority
  - Further optimization of the model could help with the precision

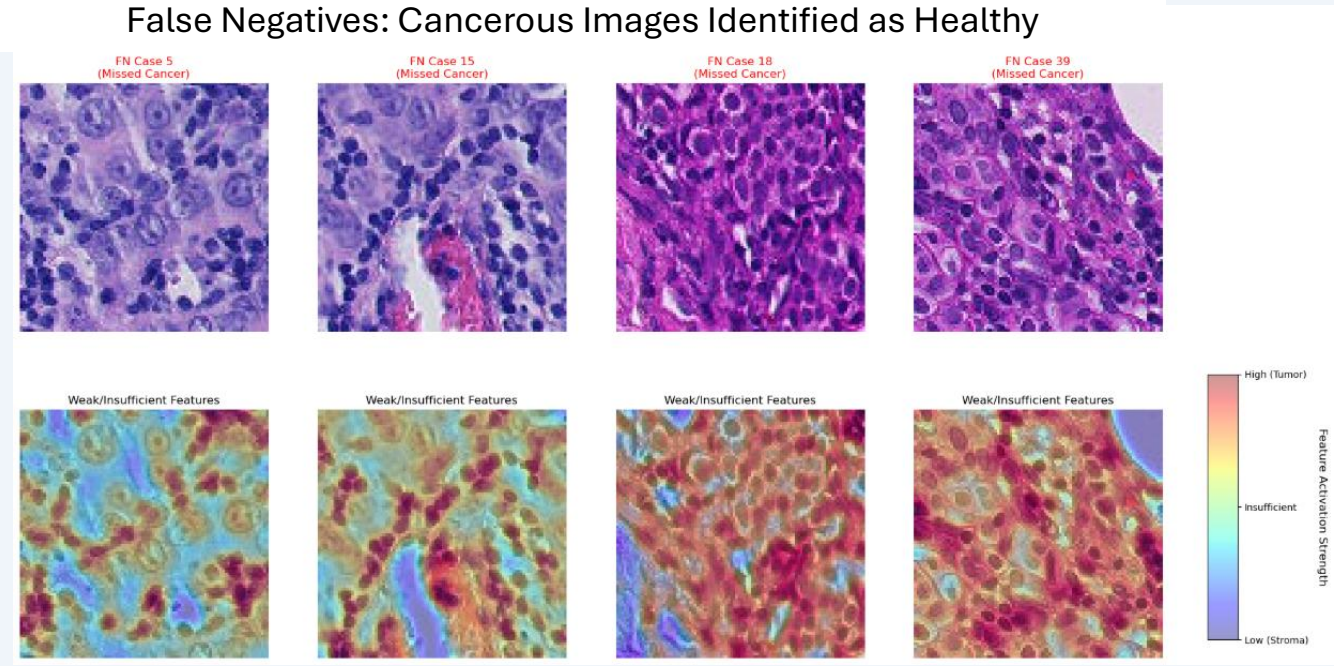


# ResNet-50: Model Feature Extraction

- When comparing the true positives to the false negatives:
  - The true positives were able to pick up on nuclear clusters well
  - The false negatives struggled to pick up on any features
    - Including oddly shaped cells

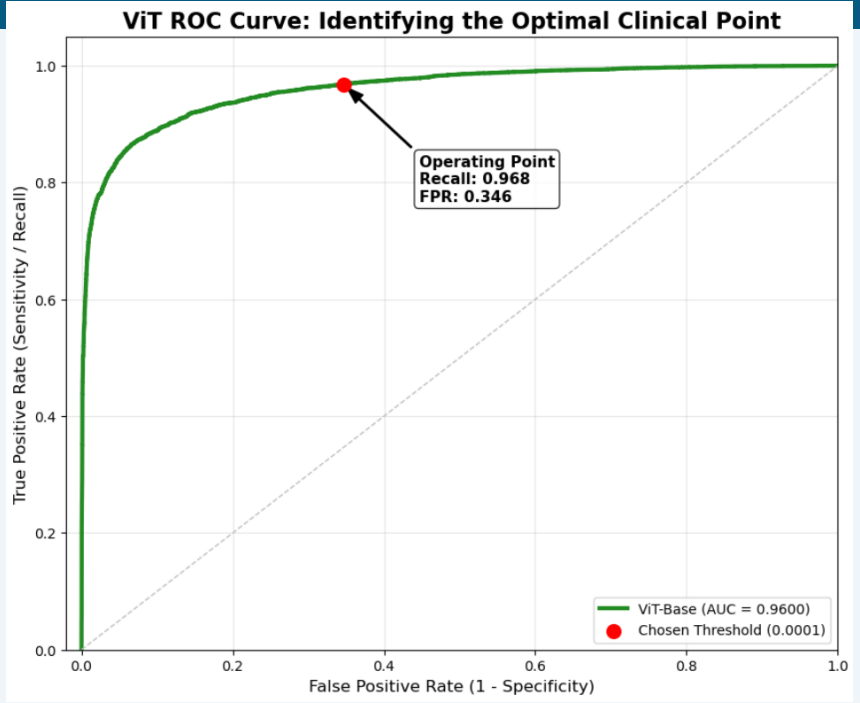


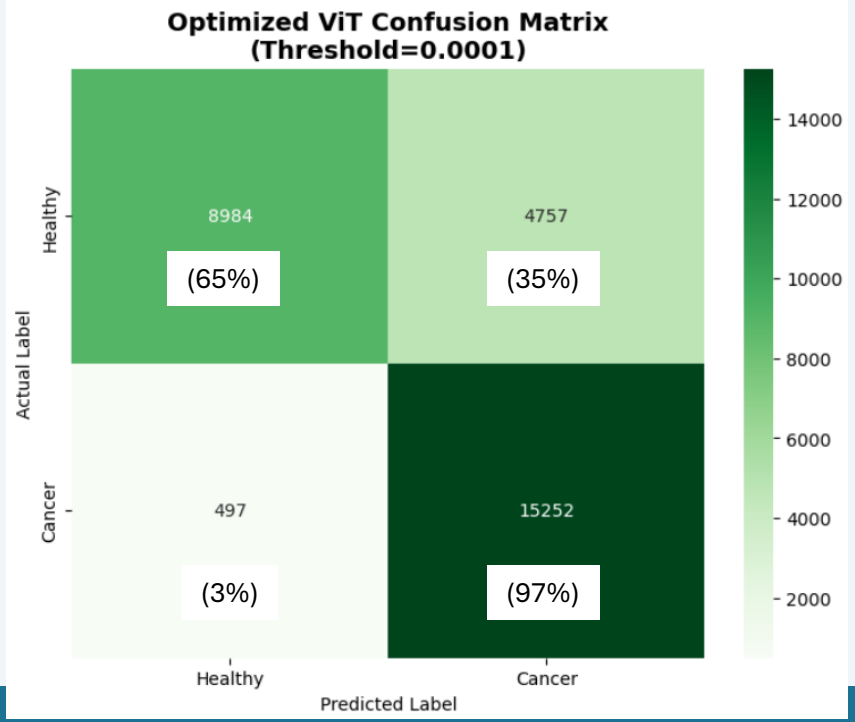



# Virtual Image Transformer (ViT)

# ViT: Final Model Performance

- The Cancer Recall was significantly improved to 97%
  - 35% of the healthy images were identified as cancerous – may lead to unnecessary treatment/testing
- This model struggles with precision, but is better than the ResNet-50 model (43%)
  - Further optimization of the model could help with the precision

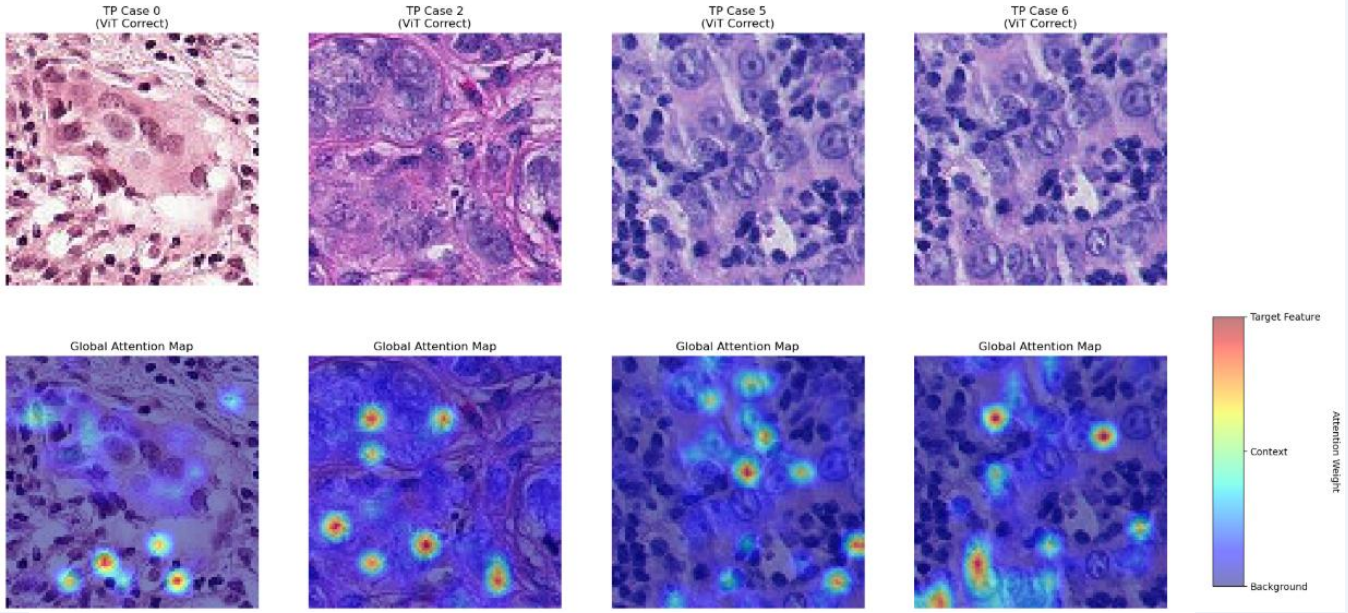


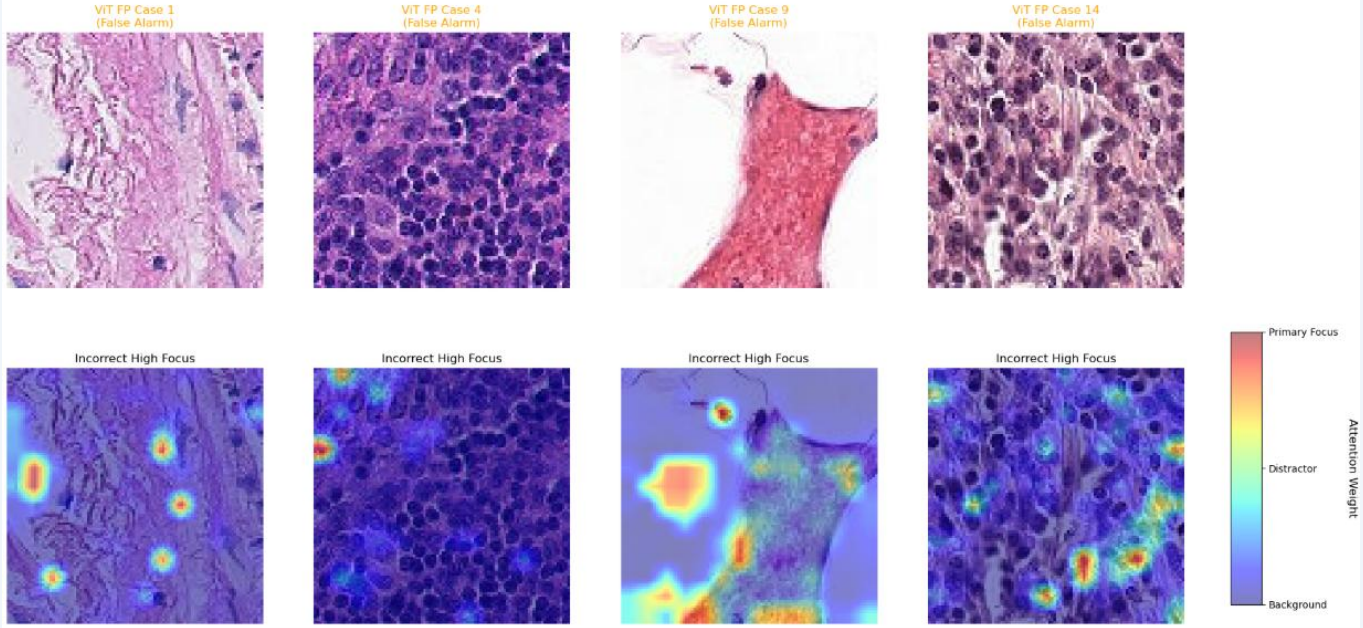
# ViT: Model Feature Extraction

- When comparing the true positives to the false negatives:
  - The true positives were able to identify pointed locations within the images
    - Ex.TP case 6. bottom left activity for strange nuclei
  - The false negatives struggled to focus on anything in particular
    - Ex. Case 9. The model focused on a white area
    - There may not have been many images like this one and so the model may have struggled with it.

True Positives: Cancerous Images




False Negatives: Cancerous Images Identified as Healthy

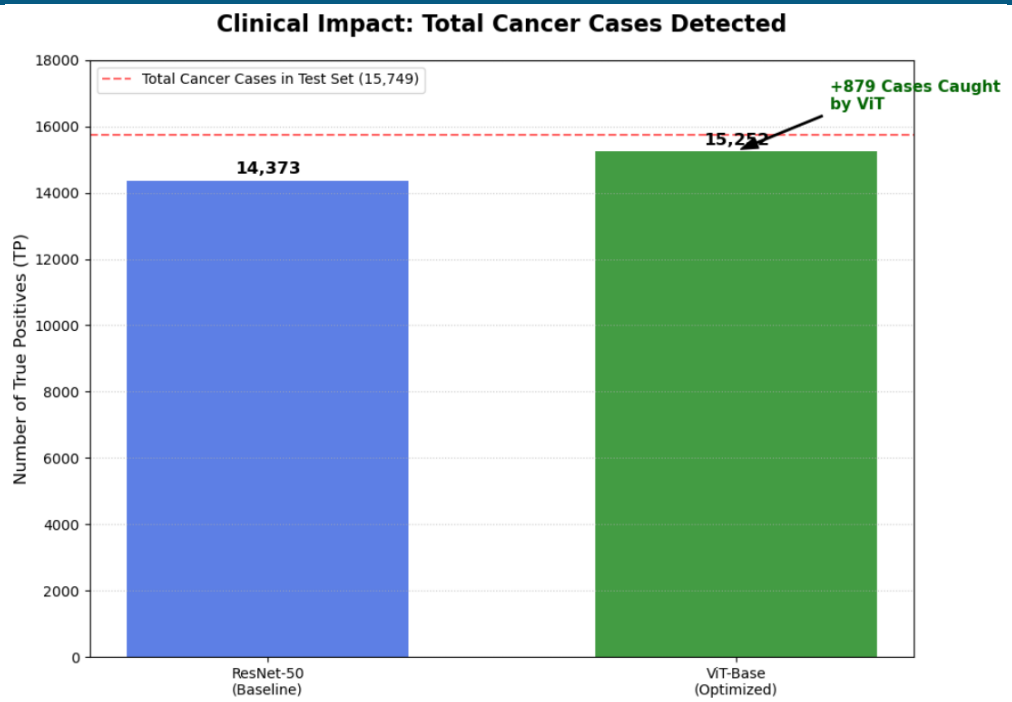


# Comparing Models

# Choosing a Model

- The ViT model was the superior model in most metrics observed
  - ResNet-50 was much fast at analyzing images

- 879 images were found to be cancerous through CiT that ResNet-50 missed




| Metric | ResNet-50 | ViT | Winning Model | Priority | Expected for Production |
|---|---|---|---|---|---|
| Recall (Cancer Cells) | 90% | 97% | ViT | 1 | >95% |
| Precision (Cancer Cells) | 71% | 76% | ViT | 2 | >95% |
| F1 Score* | 0.79 | 0.85 | ViT | 2 | >0.95 |
| Overall Accuracy | 75% | 83% | ViT | 2 | >95% |
| Time per image | 0.318 ms | 6.244 ms | ResNet-50 | 3 | <100 ms |

$$F1 = 2 * \frac{(Precision \ * \ Recall)}{Precision + Recall}$$

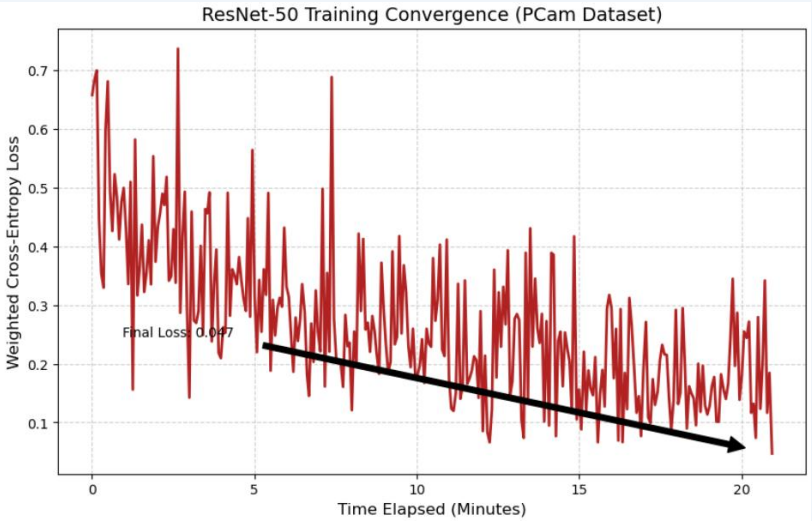
# Conclusions and Next Steps

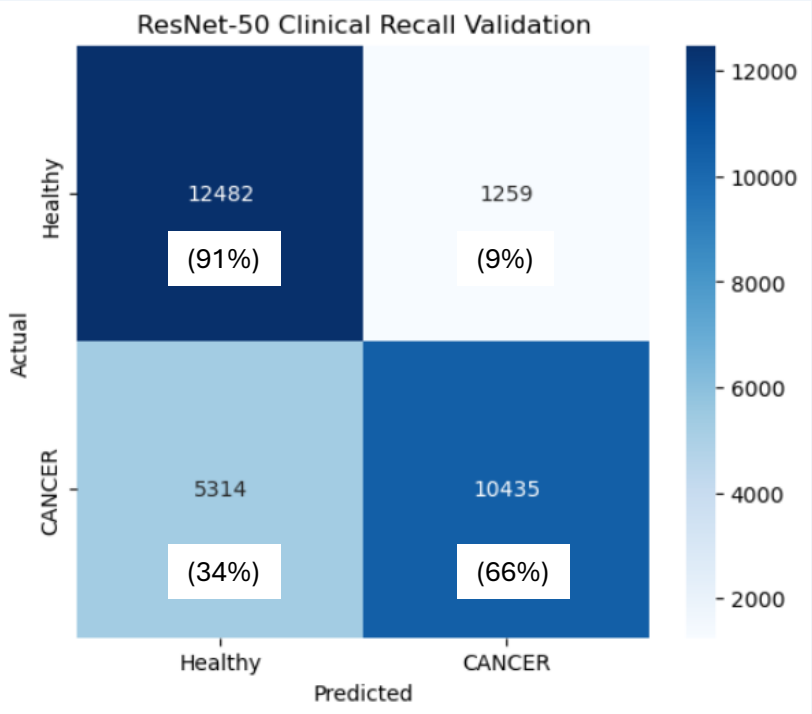
- Summary of findings:
  - Using the Patch Camelon lymphoma dataset two models were trained and compared
    - ResNet-50 was the quickest model at image classification
      - 95% faster
    - ViT was the best performer overall with the highest recall, precision, F1 score, and accuracy
      - Recall: 7% better
      - Precision: 6% better
      - F1 Score: 0.06 better
      - Overall Accuracy: 8% better
  - To get the model to deployability, it will be necessary to optimize the models further
- Next Steps:
  - Further tune parameters to search for both models
    - Ex. Learning rate, change batch sizes for training, modify contrast of images, randomly crop images to randomize cancerous location
  - Confirm best model using validation data
  - Push model towards production after further optimization

# Appendix

# ResNet-50: Initial Model Performance

- Model was trained on 960,000 images
  - ~50% of total training images
  - Decided to train with less images to limit model overfitting
- Loss function seemed to level suggesting more images may not help with training
- Recall for the Cancer images was only 66%
  - Recall for Healthy cells was 91% suggesting the model learned how to identify healthy cells well
- Next: Determine if it can be optimized by shifting confidence threshold

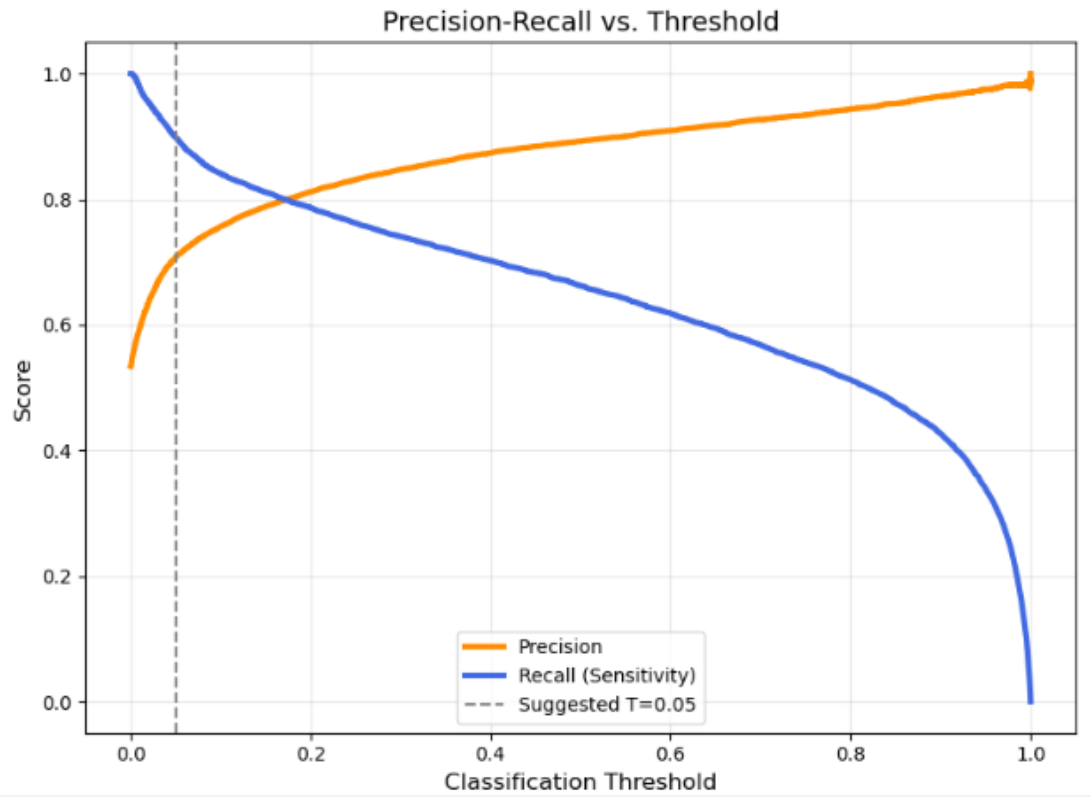


# ResNet-50: Threshold Optimization

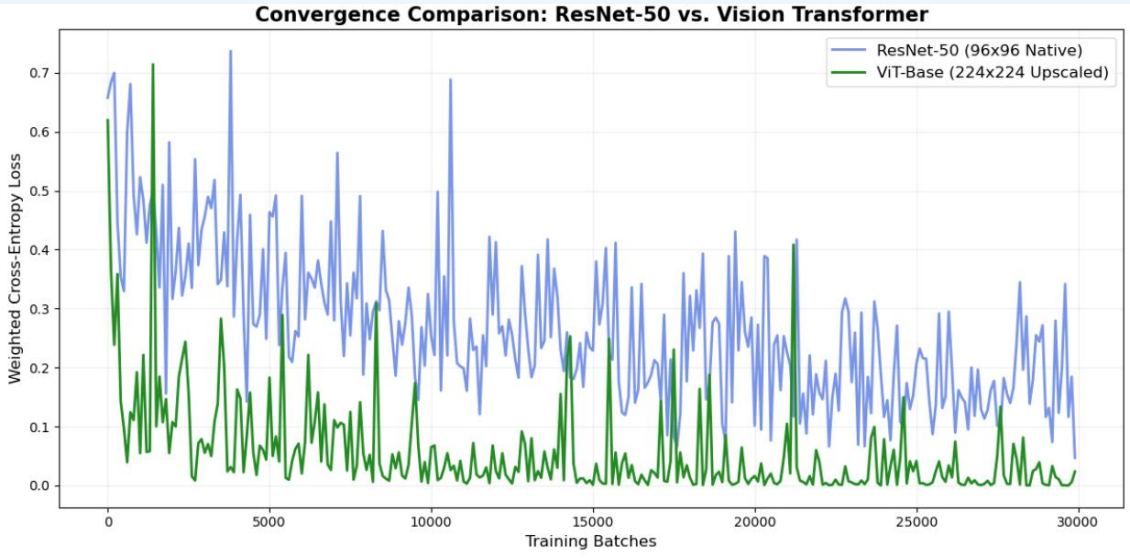
- To increase the cancer recall, a decrease in classification threshold was warranted
  - Plotted the precision and recall for a range of thresholds ranging from 1 to 0.001

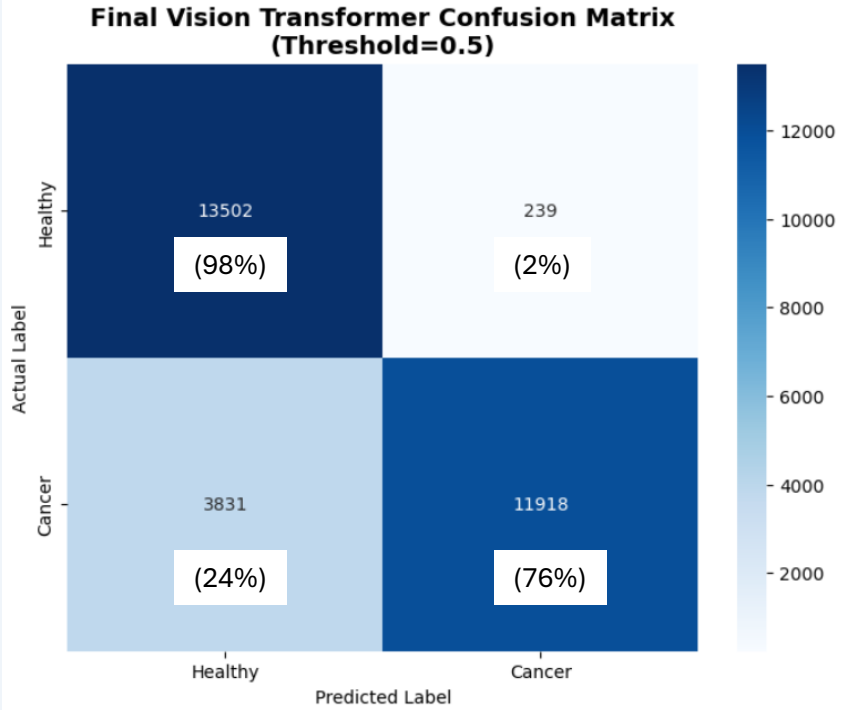
- A threshold of 0.05 was identified as hitting a 90% recall and was used moving forward
  - Precision was reduced significantly



# ViT: Initial Model Performance

- Model was trained on 960,000 images
  - ~50% of total training images
  - Decided to train with less images to limit model overfitting
- Loss function seemed to level suggesting more images may not help with training
  - Loss function reached a lower level than ResNet-50
- Recall for the Cancer images was 76%
  - Recall for Healthy cells was 98% suggesting the model learned how to identify healthy cells well
- Next: Determine if it can be optimized by shifting confidence threshold



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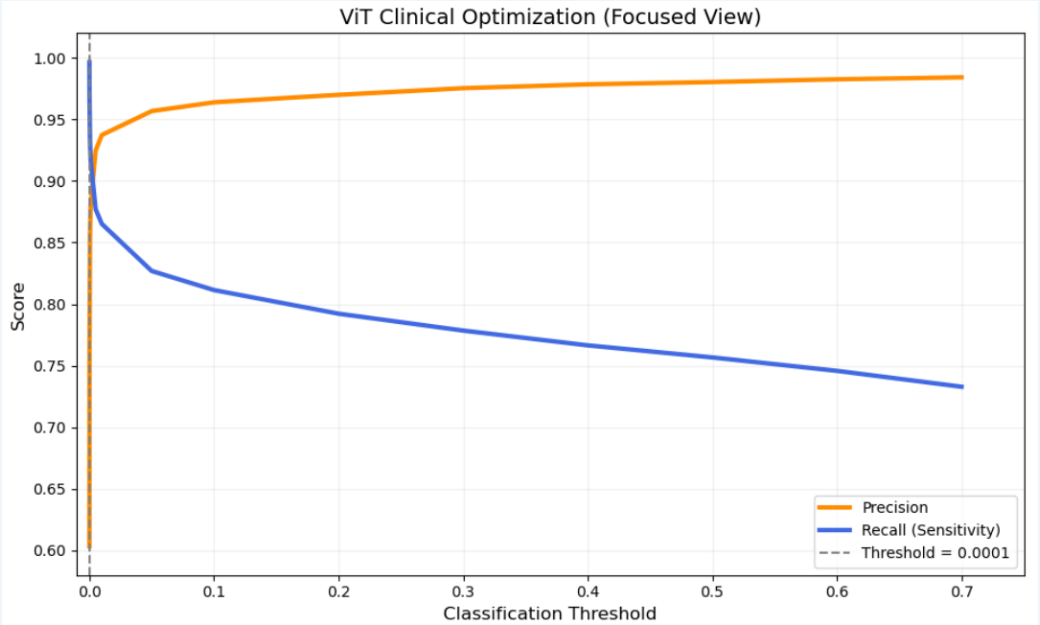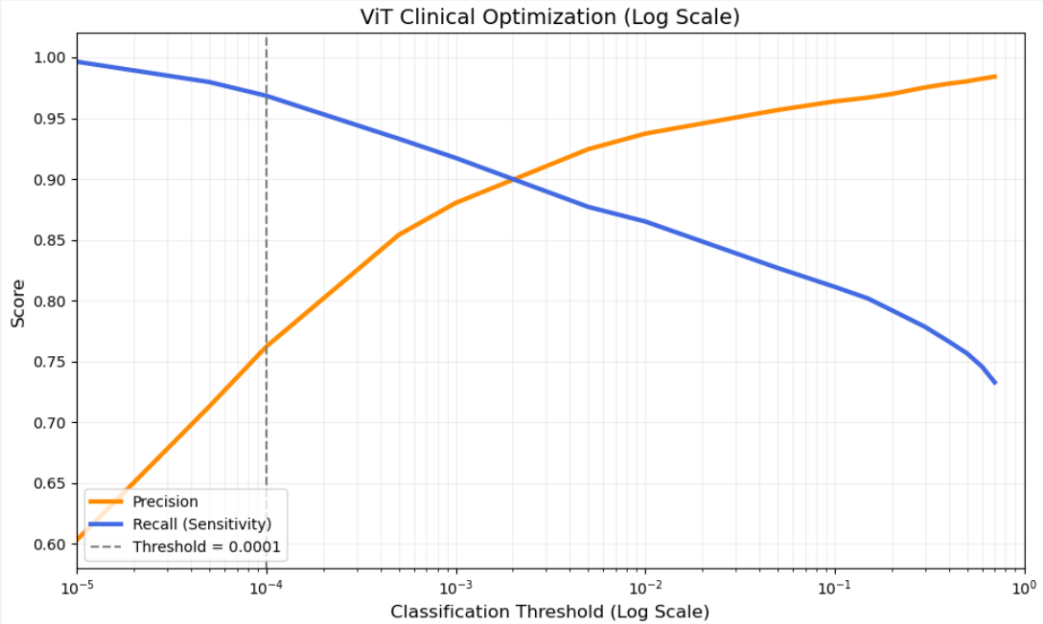
- To increase the cancer recall, a decrease in classification threshold was warranted
  - Plotted the precision and recall for a range of thresholds ranging from 0.7 to 0.00001
- A threshold of 0.0001 was identified as hitting a 97% recall and was used moving forward
  - Both recall and precision were higher for ViT than ResNet-50
  - Precision was reduced